Generative Forecasting as a Foundation Model: From Synthetic Time Series to Few-Shot Adaptation

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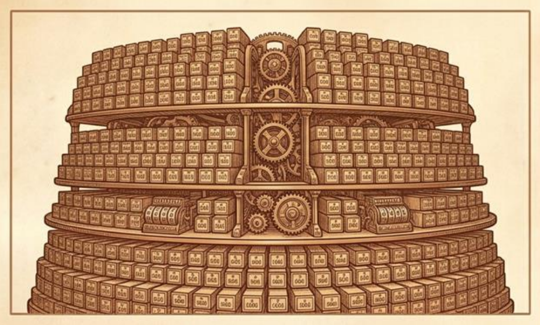
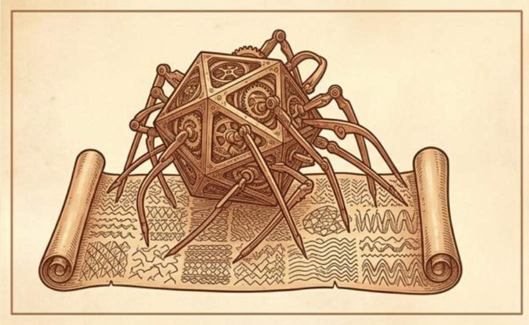
Forecasting as Generative Modeling

Forecasting at Massive Scale

Large forecasting systems operate over:



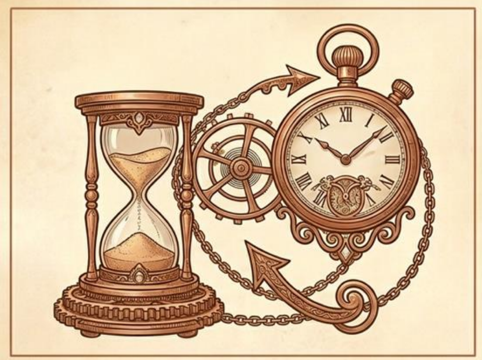
millions of SKUs



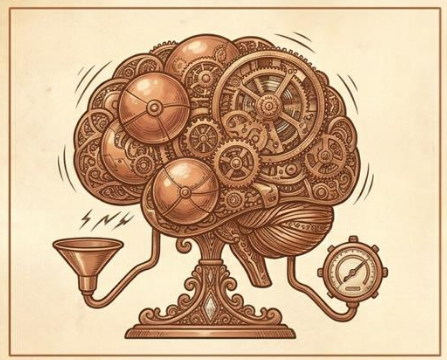
heterogeneous patterns

long-tailed dist.

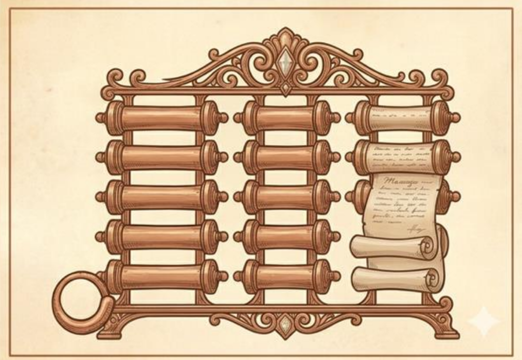
Operational constraints:



latency



inference cost



limited historical data

Toward Foundation Models for Forecasting

Foundation models are ubiquitous in:

Key ingredients:

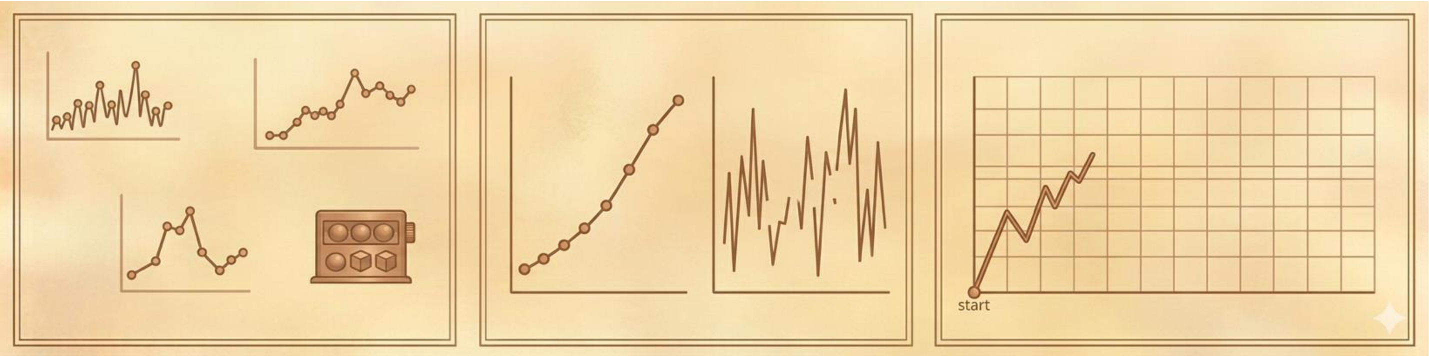
large datasets pretraining transfer across tasks

Can we build **foundation models for time-series forecasting**?

Why Foundation Models Are Hard in Time Series

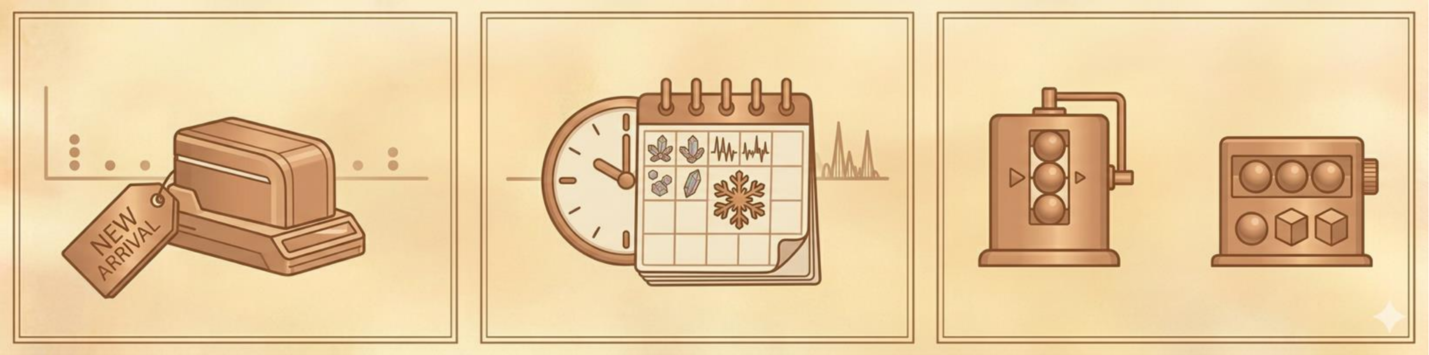
Time-series data is:

- fragmented across domains
- uneven in quality
- limited for many series



Challenges:

- cold-start SKUs
- rare regimes
- distribution shifts



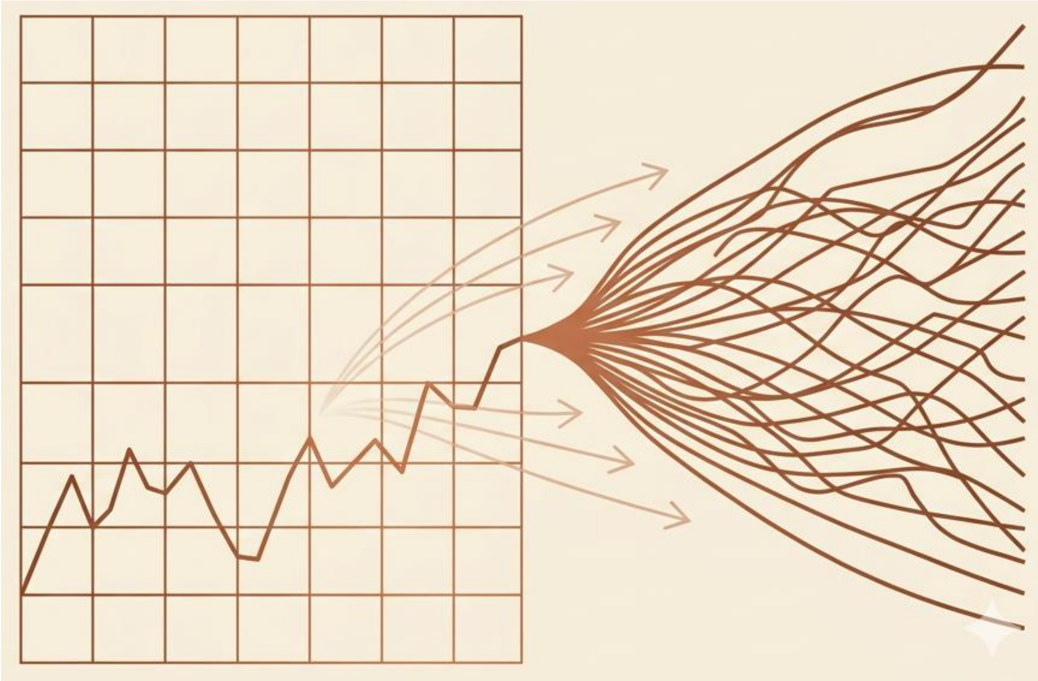
The main bottleneck may be **data coverage**, not model architecture.

Forecasting as Generative Modeling

Common view

Predict future from past

Learn the distribution



Generative view

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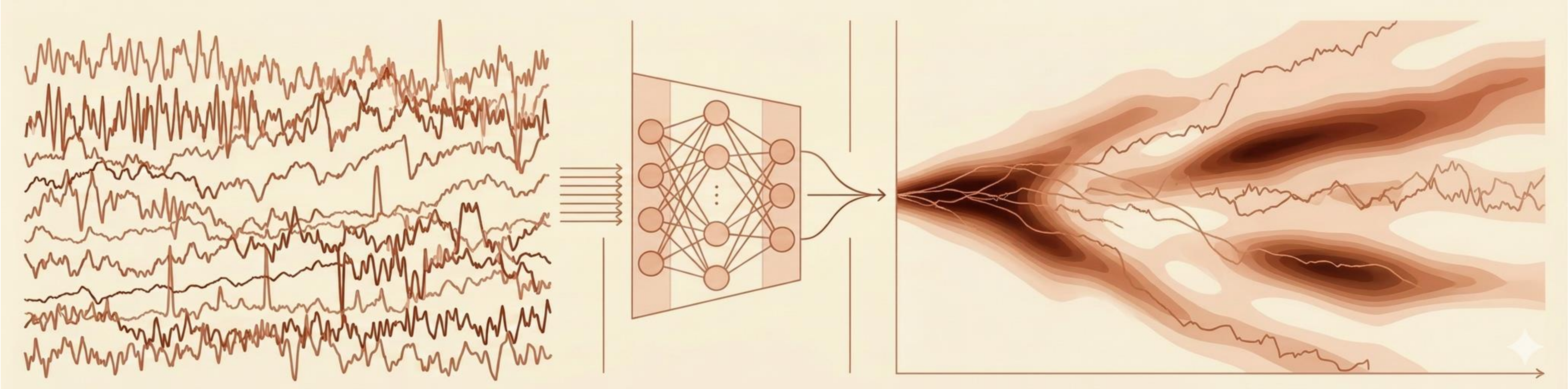
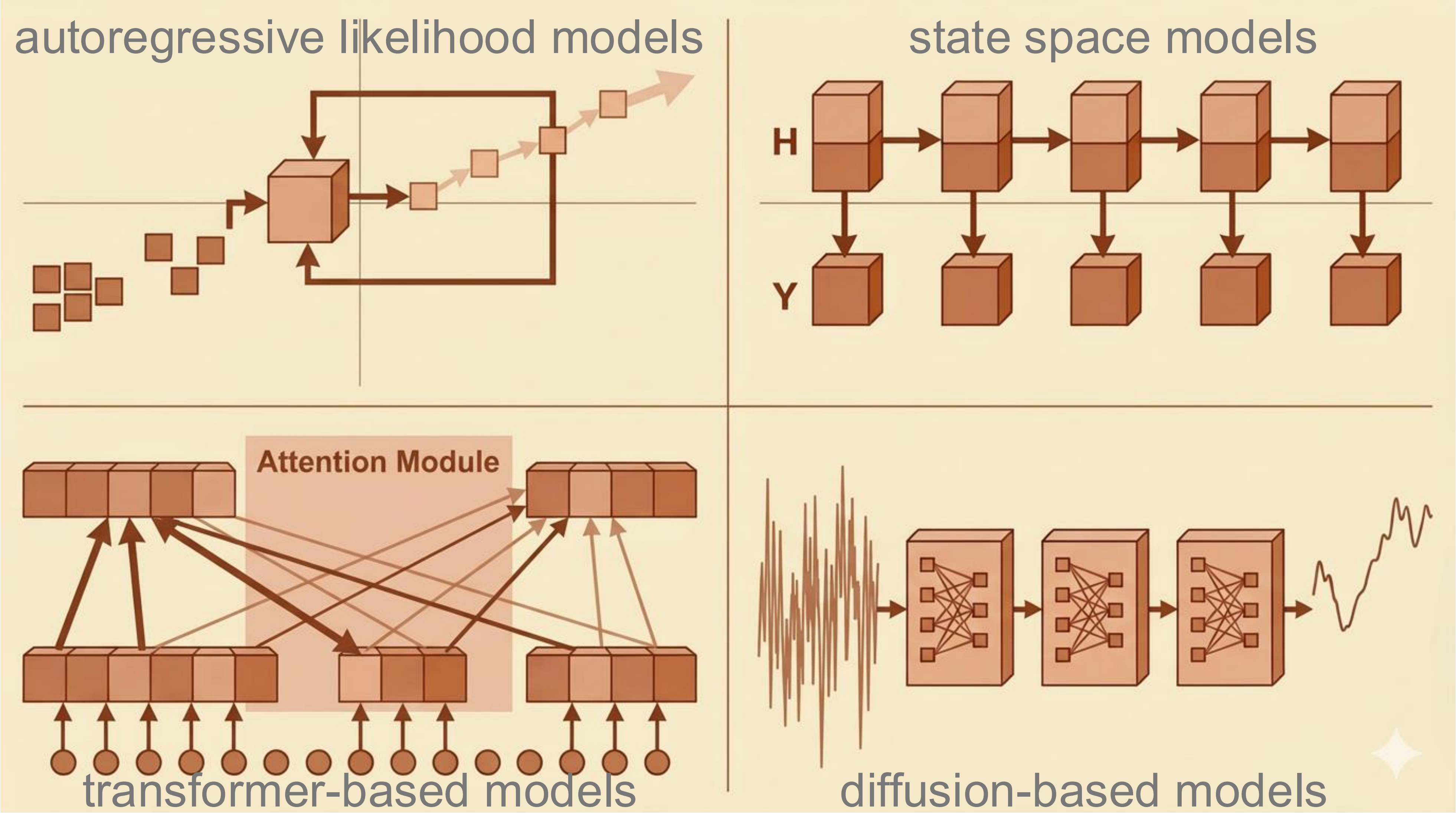
Learning the Distribution of TS Trajectories

A generative model learns the **distribution** over time-series trajectories

$$p_\theta(x_{1:T})$$



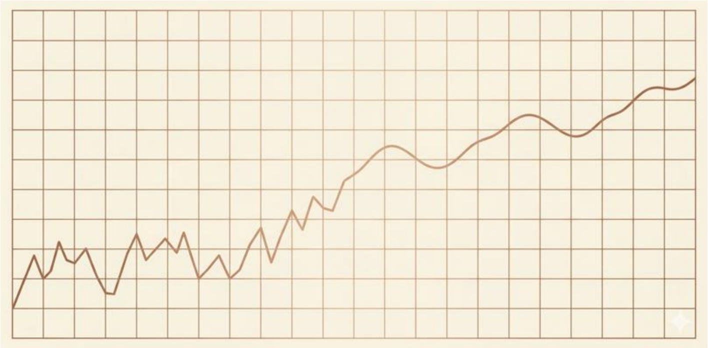
Probabilistic Forecasting Models



What the Learned Distribution Enables

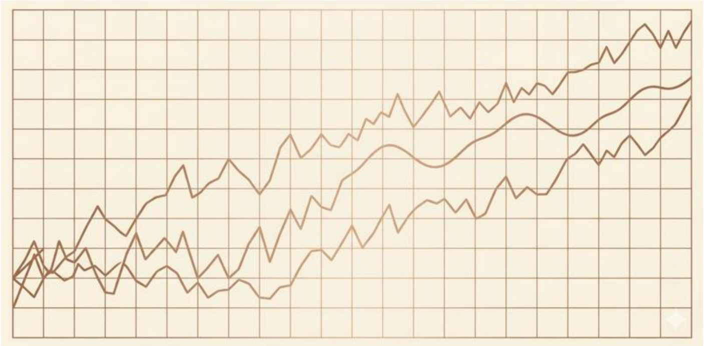
From this distribution we can obtain:

$x_{t+1:T} \sim p_\theta(x_{t+1:T} | x_{1:t})$



forecasting

$x_{t+1:T} \sim p_\theta(x_{1:T})$



trajectory simulation

$\{x_{1:T}{}^{(i)}\}_{i=1}^{N}, x^{(i)} \sim p_\theta$

synthetic datasets

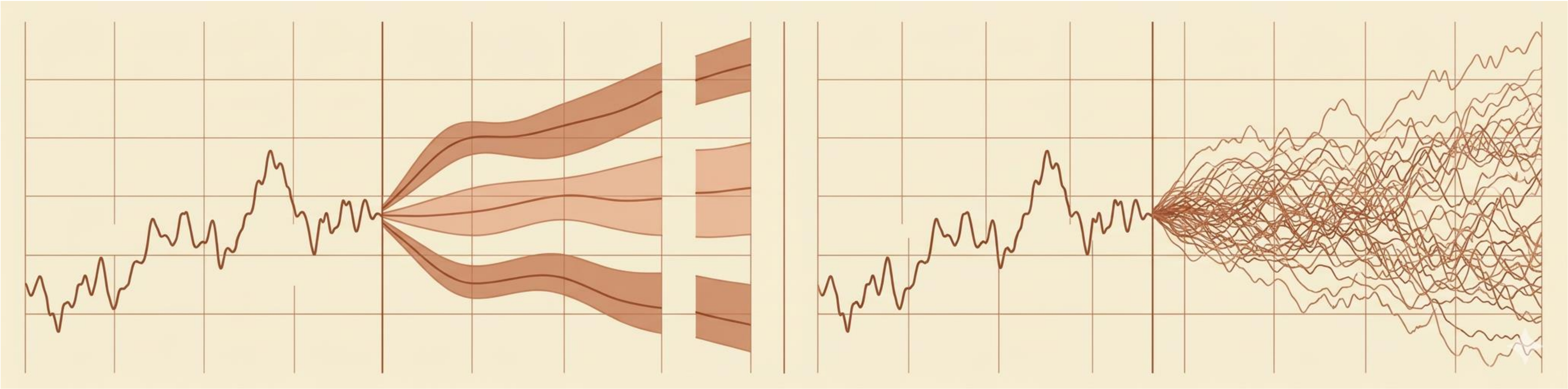
Quantile Forecasting vs Generative Forecasting

Quantile forecasting

- marginal uncertainty per horizon
- joint temporal dependence not explicit

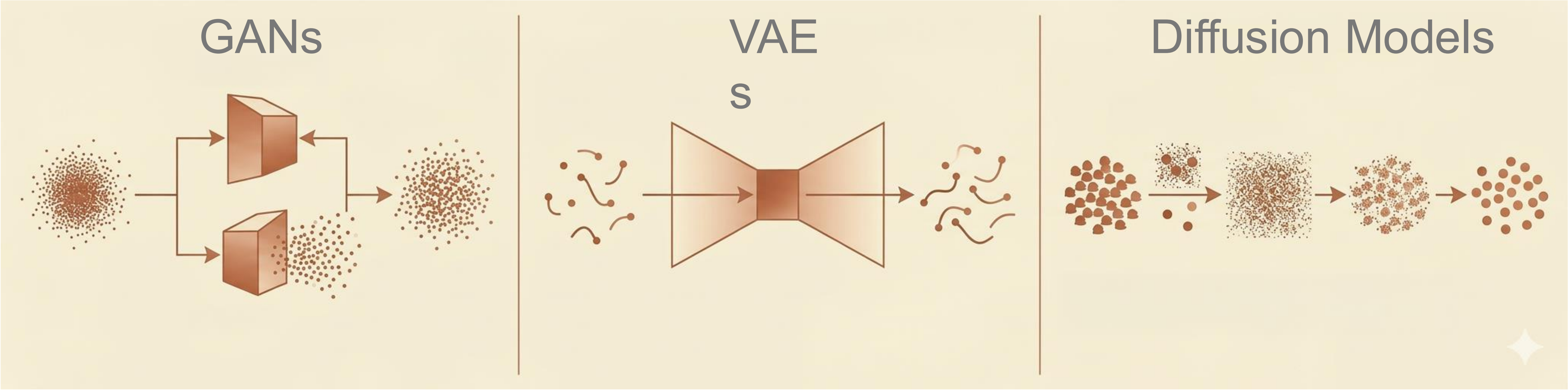
Generative forecasting

- trajectory samples
- Joint temporal structure



Learning Generative Models of Time Series

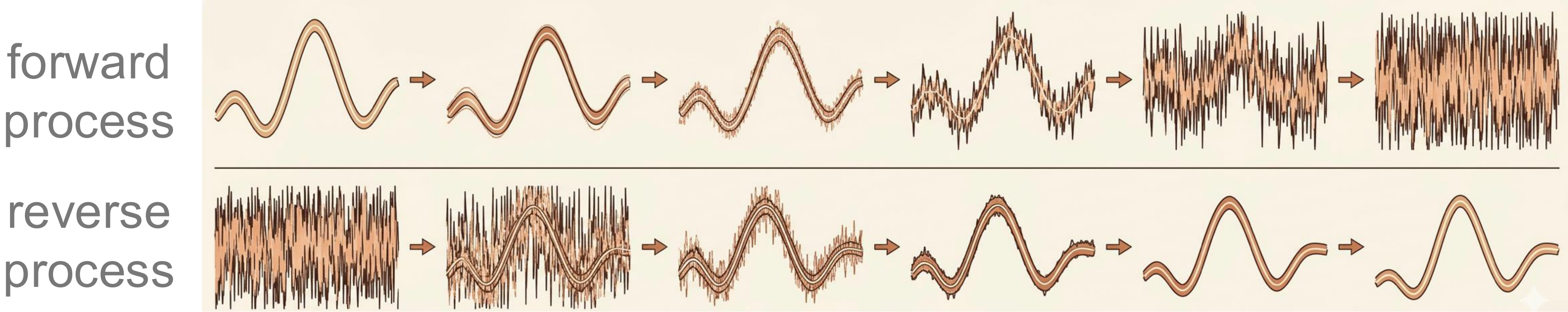
From Distributions to Generative Models



Why Diffusion Models?

Diffusion models have achieved state-of-the-art results in:

Key idea: **learn to reverse a gradual noising process**



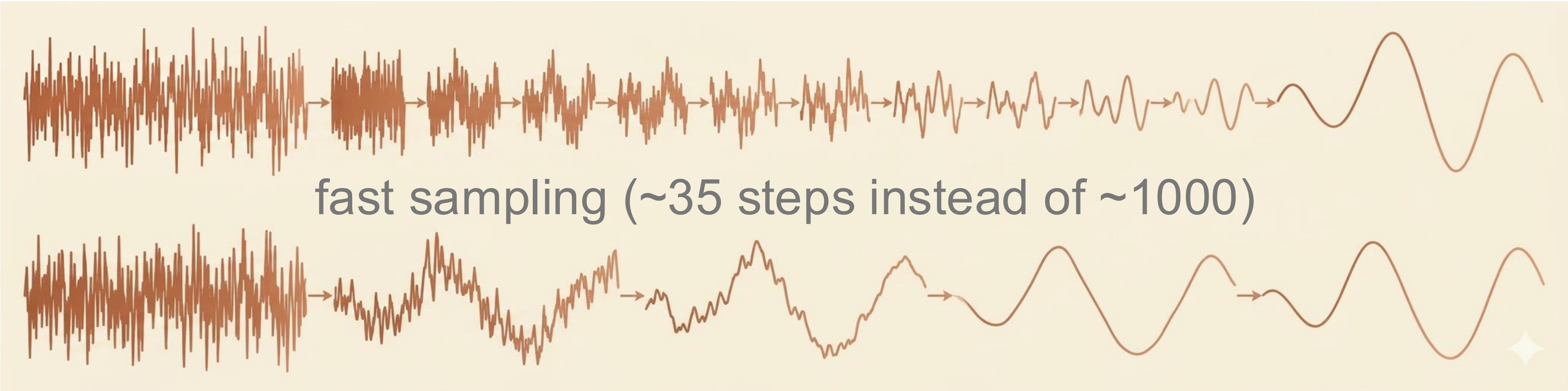
EDM: Efficient Diffusion Models

Noise perturbation

$$x_\sigma = x + \sigma\epsilon$$

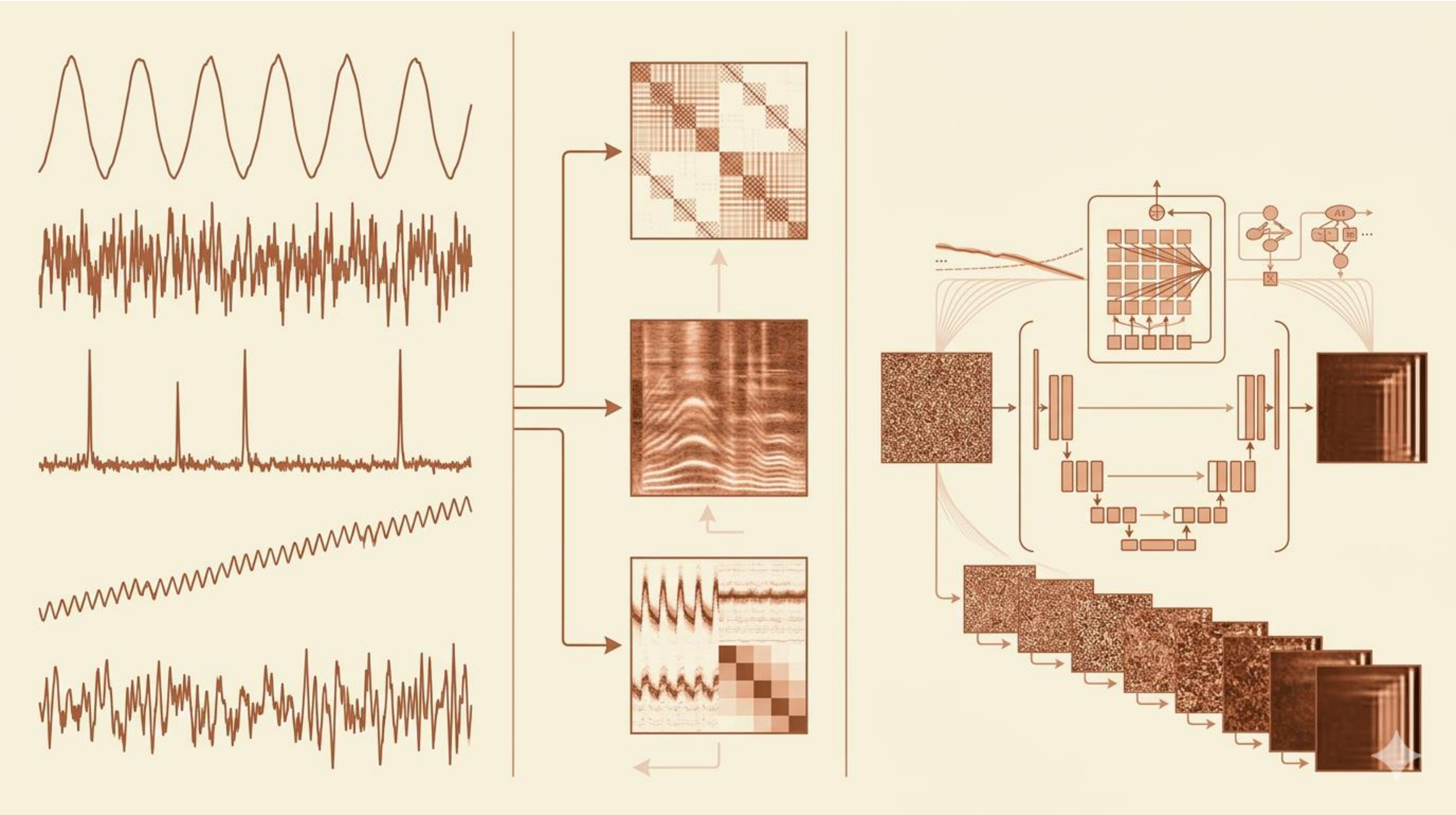
Training objective

$$\mathbb{E}_{x,\sigma,\epsilon} \, |D_\theta(x_\sigma, \sigma) - x|^2$$



Leveraging Vision Diffusion for Time Series

Most advances in diffusion models occur in **vision** (arch., training, and sampling) → Map time series into an **image representation**



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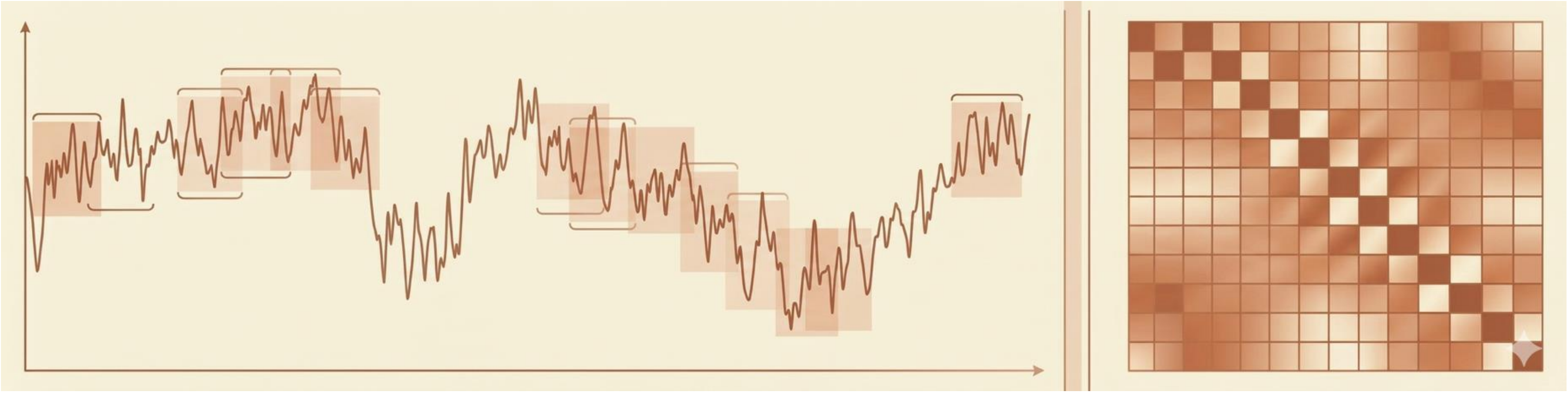
Delay Embedding: From Takens to Practice

Takens embedding: A dynamical system can be reconstructed from **delayed observations**

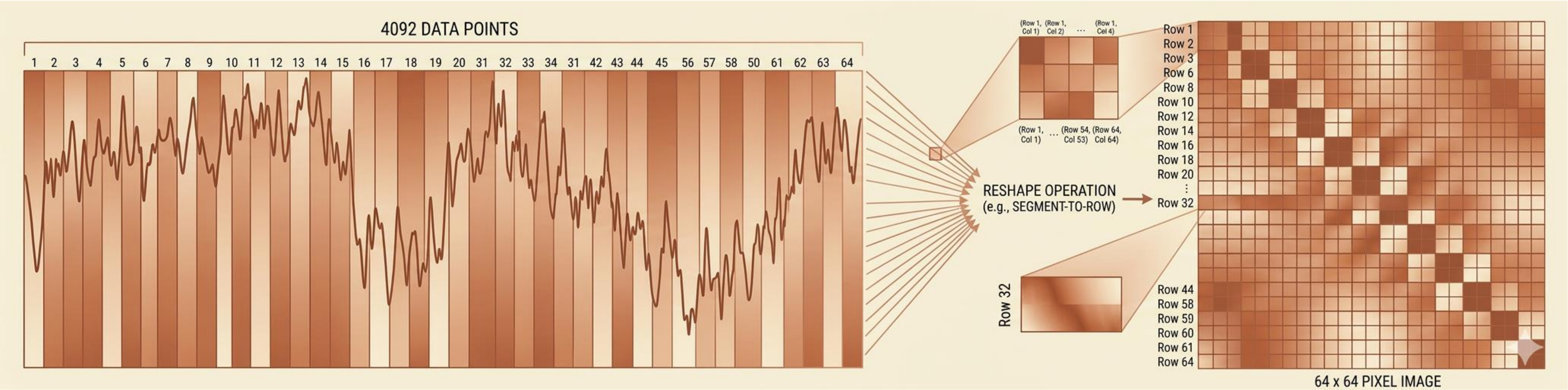
$$(x_t, x_{t+\tau}, x_{t+2\tau}, \dots, x_{t+(m-1)\tau})$$

Delay embedding in practice

$$x_{1:T} \mapsto X = \begin{bmatrix} x_1 & \cdots & x_{T-n} \\ \vdots & \ddots & \vdots \\ x_n & \cdots & x_T \end{bmatrix}$$



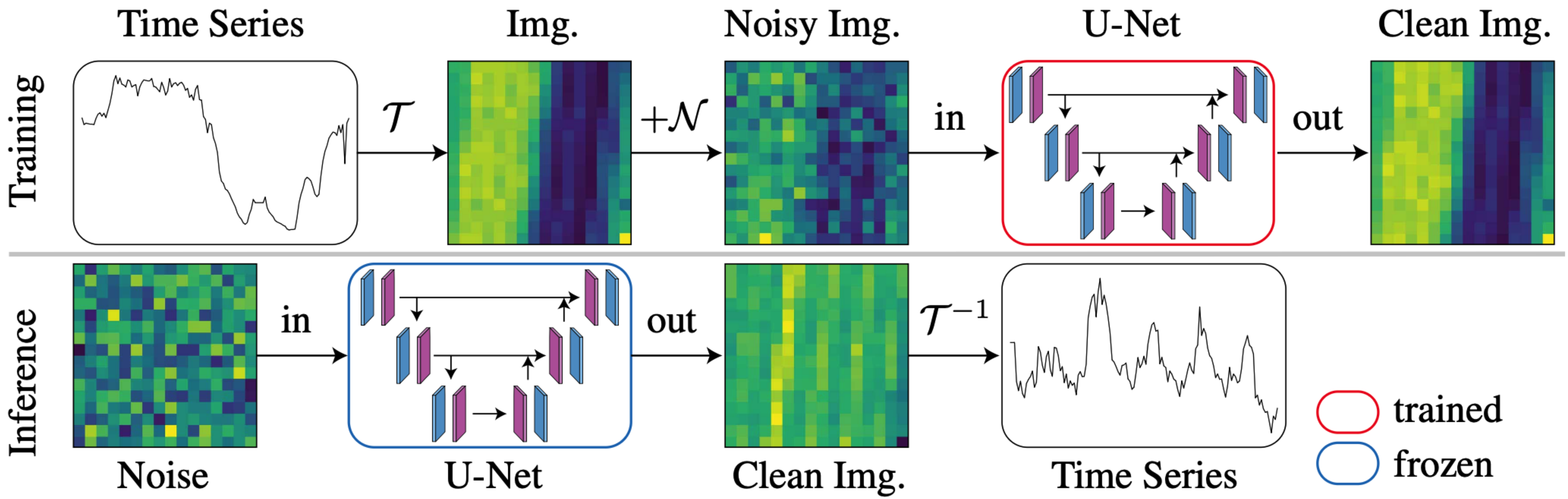
Curse of Dimensionality as an Advantage



Benefits:

- long sequences transformed to small images
- leverage powerful vision diffusion models
- Capture long-range temporal structure

ImagenTime: Diffusion Models for Time Series



Naiman, Berman, Pemper, Arbiv, Fadlon, and **A.**, NeurIPS'2024

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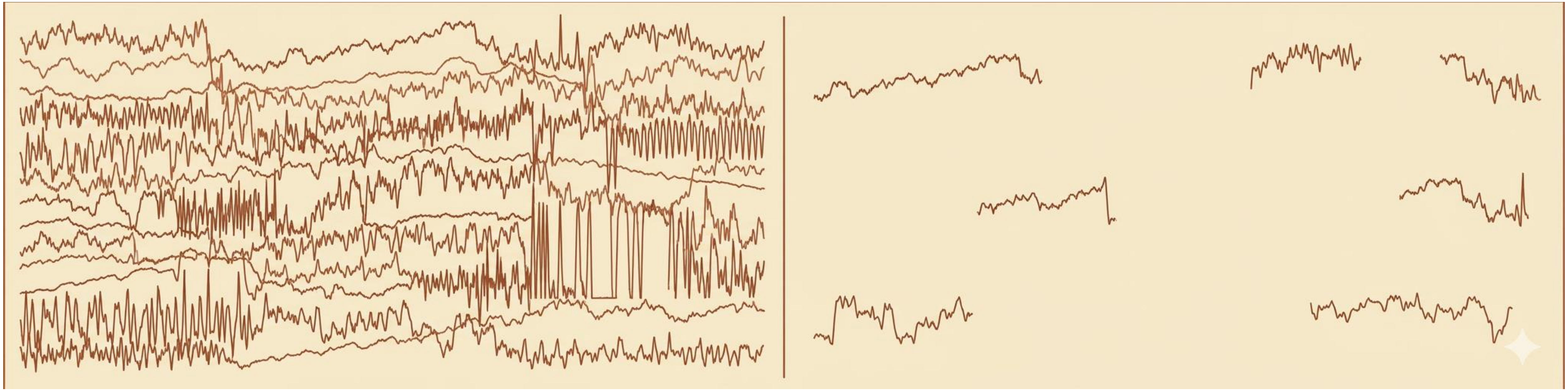
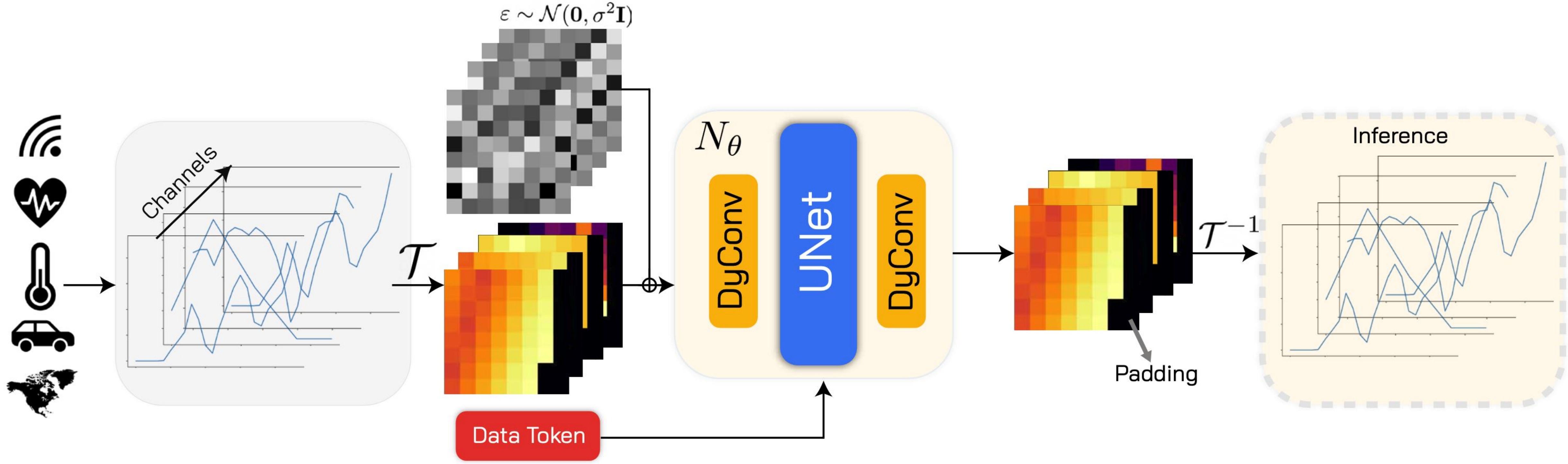
Challenge: Generative Models Require Data

Training generative models requires
- large datasets
- diverse trajectories

Time series data is often
- fragmented across domains
- limited for many series



ImagenFew: Few-Shot Modeling for Time Series



Gonen, Pemper, Naiman, Berman, and **A.**, NeurIPS'2025

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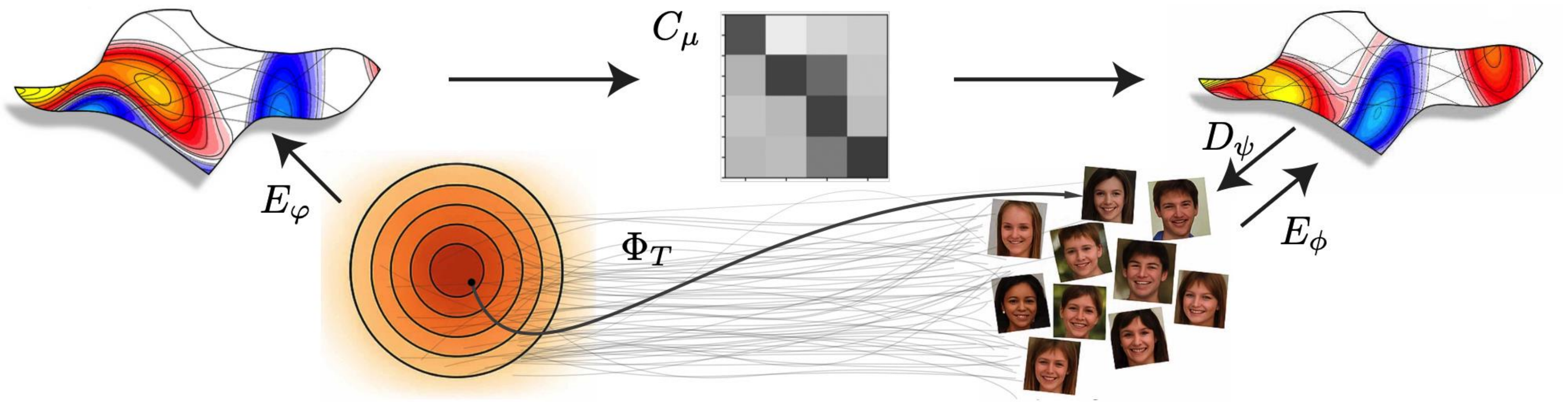
Generative Performance of ImagenTime/Few

Subset size	Metric	Ours (24)	ImagenTime	DiffTS	KoVAE	TimeGAN	Improvement
100%	Disc.	**0.027**	0.113	0.203	0.356	0.379	76.1%
	Pred.	**0.448**	0.452	0.457	0.482	0.520	0.88%
	c-FID	**0.866**	1.390	1.78	5.351	7.912	37.69%
5%	Disc.	**0.110**	0.321	0.248	0.389	0.388	55.64%
	Pred.	**0.458**	0.469	0.479	0.485	0.499	2.34%
	c-FID	**0.674**	3.464	2.087	5.232	11.055	67.7%
10%	Disc.	**0.083**	0.248	0.233	0.375	0.402	64.37%
	Pred.	**0.452**	0.458	0.469	0.491	0.494	1.31%
	c-FID.	**0.578**	2.757	1.944	5.326	14.057	70.26%
15%	Disc.	**0.066**	0.236	0.229	0.363	0.384	71.17%
	Pred.	**0.451**	0.458	0.464	0.484	0.486	1.52%
	c-FID.	**1.086**	2.211	2.064	5.644	8.290	47.38%
#10	Disc.	**0.259**	0.357	0.362	0.427	0.382	26.61%
	Pred.	0.489	0.492	0.525	0.514	**0.467**	-4.71%
	c-FID.	**3.800**	5.393	4.984	6.166	28.572	23.75%
#25	Disc.	**0.190**	0.350	0.338	0.407	0.376	43.78%
	Pred.	**0.467**	0.469	0.498	0.515	0.538	0.42%
	c-FID.	**1.582**	4.599	3.980	5.934	7.277	60.25%
#50	Disc.	**0.149**	0.342	0.313	0.397	0.367	52.39%
	Pred.	**0.460**	0.466	0.493	0.502	0.472	1.28%
	c-FID.	**0.987**	3.639	3.626	5.816	9.744	72.78%

Toward One-Step Generative Modeling

Diffusion requires many denoising steps

nonlinear dynamics $x_{t+1} = \Phi_t(x_t) \implies x_{t+1} = C_\mu x_t$ linear (Koopman) dynamics



Berman, Naiman, Eliasof, Zisling, and **A.**, NeurIPS’2025

Diffusion SDE and the Koopman View

Diffusion dynamics

$$\mathrm{d}X_t = b(X_t)\mathrm{d}t + \sigma(X_t)\mathrm{d}W_t$$

Kolmogorov backward equation

$$\partial_t u = \mathcal{L}u, \qquad \mathcal{L}f = -\frac{1}{2}\beta(t)x \cdot \nabla f + \frac{1}{2}\beta(t)\Delta f$$

Operator view: expectation evolves **linearly**

$$u(x,t) = \mathbb{E}[f(X_t)|X_0 = x]$$

Key Takeaways: Generative Models for TS

- Diffusion models learn **trajectory distributions**
- Time series can be represented as **images**
- Generative priors enable **few-shot modeling** and **single-step sampling**

From Generation to Forecasting

Key Observations:

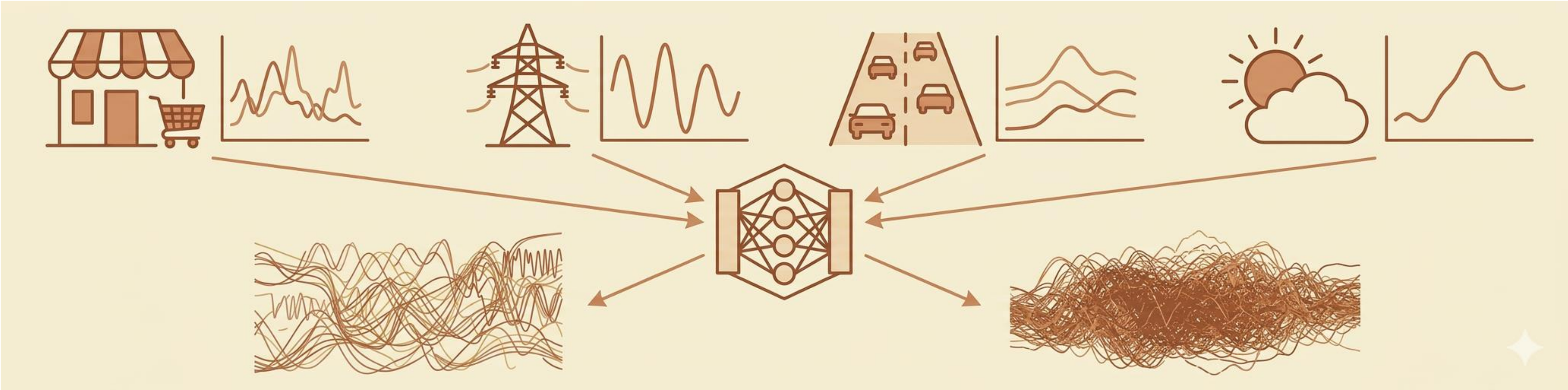
- Forecasting performed via **imputation in diffusion space**
- Preliminary results **without training** a dedicated forecasting model

Dataset	Extrapolation				
	ODE-RNN	Latent ODE	CRU	LS4	Ours
ETTh1	-	1.02	1.02	3.42	**.701**
ETTh2	-	1.17	1.09	3.83	**.667**
ETTm1	-	**.592**	.643	3.05	.634
ETTm2	-	.414	.378	3.64	**.348**
Physionet	3.01	4.21	6.29	4.94	**1.34**
USHCN	1.96	2.03	1.27	4.36	**1.20**
Traffic	-	1.01	*	2.23	**.221**
KDD-Cup	-	.696	.723	6.51	**.368**

Generative Foundation Models for Forecasting

Designing Foundation Models for Forecasting

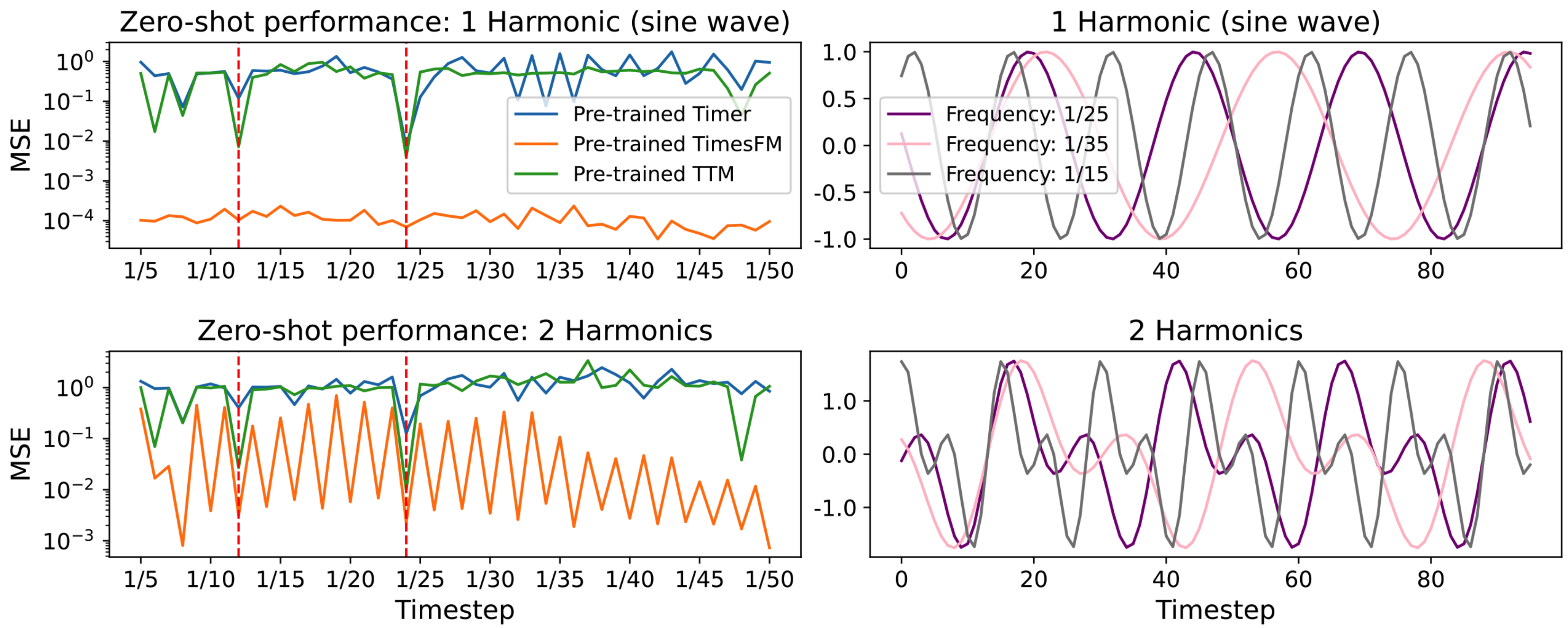
Goal: Train a single model that generalizes across many TS domains



Generative models can **unify** time series modeling across domains

Frequency Coverage in Forecasting Data

Foundation forecasting models often learn **spectral biases** from their training data, leading to **overfitting** dominant Fourier components



Nochumsohn, Moshkovitz, Avner, Di Castro, and **A.**, arXiv’2024

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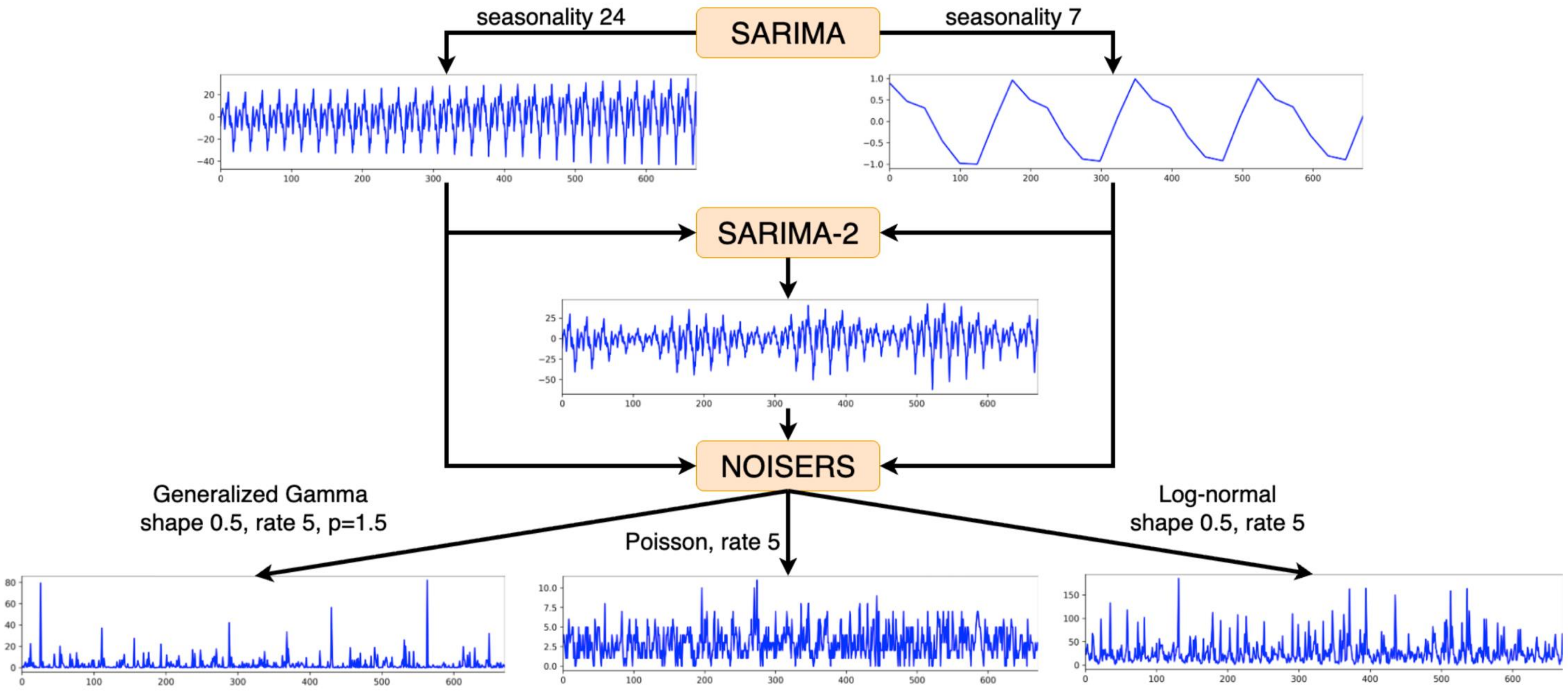
Foundation Forecasting with Synthetic Data

Foundation forecasting models require **very large training datasets**, motivating training on **fully synthetic** datasets



Oreshkin et. al, ICLR'2026

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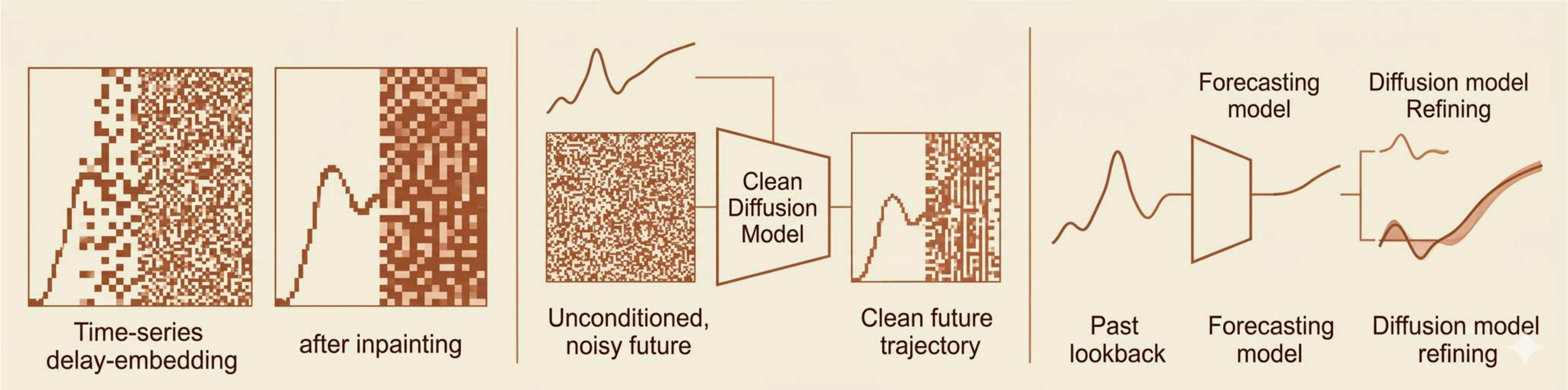
Extending ImagenFew for Forecasting

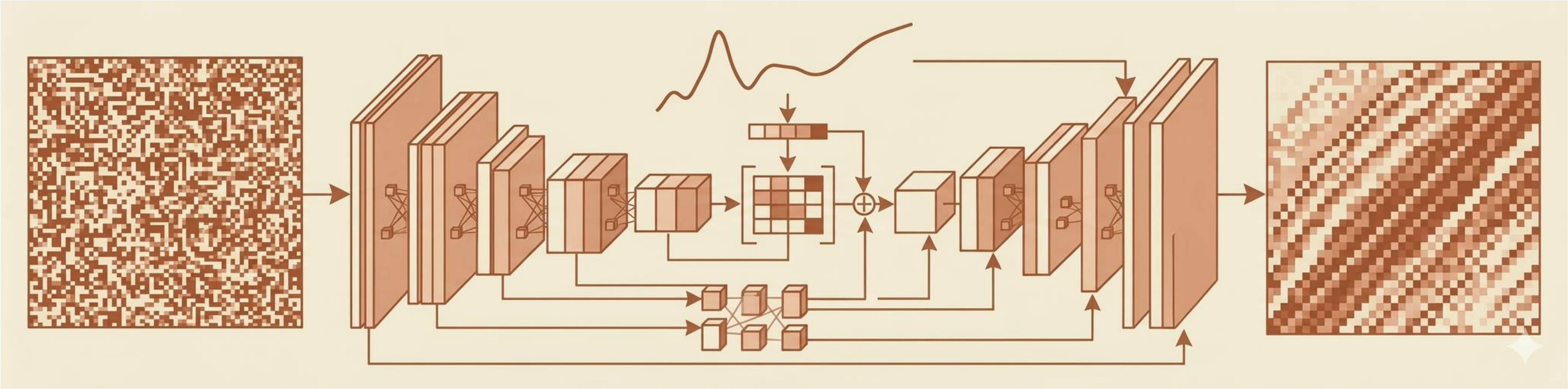
inpainting | future diffusion | residual denoising



Structured Forecasting with Koopman Dynamics

Combine **future diffusion** with **structured dynamics**:

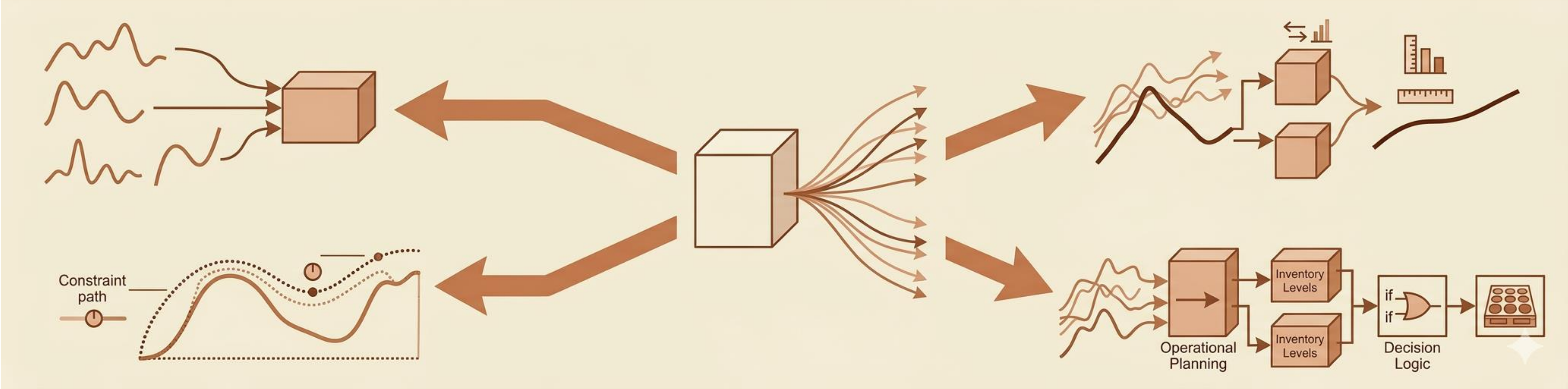
- enforce temporal consistency
- introduce interpretable linear dynamics
- guide generative forecasting



Open Challenges

Key research directions:

- scalable foundation models for forecasting
- controllable generative models
- evaluation of trajectory forecasts
- integrating generative models with decision systems



Takeaways

Forecasting can be framed as **generative modeling**.

Diffusion models enable **learning trajectory distributions**.

Generative models may be the backbone of **future forecasting systems**.

Thank you!

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